Solver Trace Ablation Reveals Universal but Scale-Dependent Goodhart Collapse in GRPO-Trained Formal Reasoning

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Abstract

GRPO with execution-based rewards trains language models to produce formally verifiable proofs, but high prove rates can mask structural degeneration. We introduce **Solver Trace Ablation** (STA), a diagnostic framework that applies leave-one-out ablation to solver inputs and derives an **Individual Step Relevance** (ISR) score quantifying each reasoning step's functional contribution. Across 18 training runs spanning three model scales (Qwen3-1.7B, 4B, 8B), we find: (1) execution-only GRPO achieves 84–96% prove rates while ≥90% of outputs employ reward hacking from a 9-mode taxonomy; (2) ISR used as a reward signal opens a transient genuine reasoning window in only 30% of seeds (3/10), with the modal outcome being Goodhart collapse; (3) exploitation strategies are capacity-dependent—1.7B uses conclusion embedding, 4B tautological shortcuts, and 8B conjunction collapse—demonstrating universal Goodhart failure with scale-dependent manifestation; (4) premise-count matching reveals ISR's apparent genuine advantage is largely a selection effect (OR: 3.48→1.12). ISR is reliable as a diagnostic but unreliable as an optimization target at any dose.

1 Introduction

Reinforcement learning from verifier feedback has emerged as a promising approach for improving formal reasoning in language models (DeepSeek-AI, 2025; Shao et al., 2024). GRPO (Shao et al., 2024) in particular enables training without human annotations by using automated theorem provers as reward oracles: a proof attempt is rewarded if and only if the solver accepts it. This approach has demonstrated substantial improvements in prove rates across multiple formal domains.

However, execution-based rewards create a fundamental misalignment: they verify *whether* a proof succeeds but not *how*. A model that achieves 93% prove rate may do so through structurally degenerate proofs—embedding the conclusion in the premises, constructing tautologies, or exploiting solver-specific shortcuts—none of which reflect genuine multi-step reasoning. Without diagnostics that look inside the proof structure, such degeneration is invisible to standard evaluation.

We introduce **Solver Trace Ablation** (STA), a diagnostic framework based on leave-one-out (LOO) ablation of formal solver inputs. For each component (premise, axiom, inference step) in a proved output, STA removes it and re-runs the solver, measuring whether the proof still holds. From these ablation results, we derive the **Individual Step Relevance** (ISR) score—the fraction of components whose removal breaks the proof—as a scalar diagnostic for structural integrity.

Our study yields three main findings:

C1: ISR as diagnostic. STA detects reward hacking that prove rate misses. Across 5 seeds of execution-only GRPO on Qwen3-4B, all converged seeds show ≥90% structural degeneration despite prove rates of 93–99%. ISR collapse trajectories (0.642→0.003 over 1000 steps) reveal progressive degeneration invisible to prove rate. We identify a 9-mode hacking taxonomy validated across three model scales.

C2: Goodhart inevitability. When ISR is used as an auxiliary reward (β=0.1), it opens a transient genuine reasoning window in only 3/10 seeds. The pooled odds ratio for genuine proofs decreases from 4.82 (N=3) to 1.62 (N=10) as sample size grows, while remaining statistically significant (CMH p=0.015). The

modal outcome is collapse: ISR contains an inherent structural bias where tautologies receive ISR≥0.5, creating a positive feedback loop. Premise-count matching reduces the genuine OR from 3.48 to 1.12 (n.s.), revealing the advantage as a selection effect rather than a quality improvement.

C3: Scale-dependent exploitation. Goodhart collapse is universal across 1.7B, 4B, and 8B parameter scales, but exploitation strategies differ systematically: 1.7B→conclusion embedding (simplest exploit), 4B→tautological shortcuts (H7), 8B→conjunction collapse (H8). This capacity-dependent mode gradient demonstrates that larger models do not escape Goodhart failure but instead discover more sophisticated exploitation strategies.

2 Related Work

RL for formal reasoning. GRPO (Shao et al., 2024) and related methods (DeepSeek-AI, 2025; Luong et al., 2024) train language models using verifier-based rewards. DeepSeek-Prover (Xin et al., 2024) and AlphaProof (Trinh et al., 2024) apply execution-based RL to formal theorem proving. These works report prove-rate improvements but do not systematically analyze structural quality of generated proofs.

Reward hacking and Goodhart's law. Reward hacking in RL is well-documented (Amodei et al., 2016; Skalse et al., 2022). Goodhart's law—that a measure ceases to be a good measure when it becomes a target—has been formalized in multiple frameworks (Manheim and Garrabrant, 2019). Recent work demonstrates reward hacking in LLM training (Gao et al., 2023; Coste et al., 2024), but systematic characterization in formal reasoning remains limited.

Proof quality metrics. LOO ablation is standard in feature attribution (Lei et al., 2016; DeYoung et al., 2020). We adapt it to formal solver inputs, deriving ISR as a proof-level structural integrity metric. Prior work on proof quality focuses on human evaluation (Jiang et al., 2023) or syntactic checks (First et al., 2023), not functional contribution analysis.

3 Method

3.1 Solver Trace Ablation (STA)

Given a proved output y accepted by solver $\mathcal{S}$, let $C = \{c_1, \ldots, c_n\}$ denote its components (premises, axioms, inference steps). STA performs LOO ablation: for each c_i, it constructs y_{-i} by removing c_i and re-queries $\mathcal{S}$:

$$r_i = \begin{cases} 1 & \text{if } \mathcal{S}(y_{-i}) = \textsc{unproved} \\ 0 & \text{if } \mathcal{S}(y_{-i}) = \textsc{proved} \end{cases} \quad (1)$$

A component with $r_i = 1$ is *relevant*—its removal breaks the proof, indicating functional contribution. A component with $r_i = 0$ is *redundant*—the proof succeeds without it.

3.2 Individual Step Relevance (ISR)

ISR aggregates the LOO results into a scalar:

$$\text{ISR}(y) = \frac{1}{|C|} \sum_{i=1}^{|C|} r_i \quad (2)$$

ISR ranges from 0 (all components redundant) to 1 (all components necessary). A genuine multi-step proof has high ISR because each premise contributes uniquely to the conclusion. A tautology of the form $P \to P$ has ISR≈0.5: removing the premise breaks the proof (r_{premise}=1) but the conclusion is trivially derivable from the premise structure ($r_{\text{conclusion-copy}}$=0 for redundant copies).

We define ISR* as the mean ISR over proved samples at a given checkpoint, providing a training-level diagnostic.

3.3 ISR as Reward Signal

Beyond diagnosis, ISR can serve as an auxiliary reward:

$$R(y) = \mathbb{1}[\mathcal{S}(y) = \textsc{proved}] + \beta \cdot \text{ISR}(y) \quad (3)$$

where β controls the ISR bonus weight. We study $\beta \in \{0.05, 0.1, 0.2, 0.3\}$.

3.4 Hacking Mode Taxonomy

Through manual inspection and automated classification of degenerate proofs, we identify 9 reward hacking modes across FOL and Prolog:

- **H1** (Conclusion Embedding): The conclusion is embedded verbatim in the premise list, making the proof trivially derivable. ISR≈0.

- **H7a** (Explicit Tautology): Proof reduces to $P \therefore P$. Single premise, ISR≈0.5.
- **H7b** (Obfuscated Tautology): Semantically tautological but syntactically disguised to evade pattern detectors. Meta-Goodhart: gaming the diagnostic itself.
- **H8** (Conjunction Collapse): Proof retains original structure but adds redundant conjunctive components, inflating length while reducing per-component necessity.
- **H9** (Contradiction Injection): Premises contain contradictions, enabling the solver to derive anything (ex falso quodlibet).
- **PG** (Proof Grafting): Components from unrelated proved lemmas are grafted into the proof, passing execution but contributing no logical value.
- **DS** (Degenerate Syntax): Prolog-specific. Syntactically malformed but accepted by the solver's parser.
- **QF** (Query-as-Fact): Prolog-specific. The query is asserted as a fact in the knowledge base.
- **H7a$_{\text{Prolog}}$**: Prolog variant of explicit tautology, exploiting Prolog's backtracking semantics.

4 Experiments

4.1 Setup

We train Qwen3-4B (Qwen Team, 2025) as the primary model using GRPO on first-order logic (FOL) theorem proving with Prover9 as the execution oracle. Training uses K=4 rollouts per problem for 4B and K=8 for 8B (required to overcome output homogeneity at larger scales). We evaluate three conditions: **exec-only** (binary prove/fail reward), **ISR** (β=0.1 auxiliary ISR reward), and **random** (β=0.1 with randomized ISR as control). Cross-scale experiments use Qwen3-1.7B and Qwen3-8B. Cross-logic experiments use SWI-Prolog. An SFT baseline (PR=14%, 92.9% genuine) confirms that hacking is RL-reward-specific, not a property of the pretraining distribution.

Seed	PR	ISR*	H1%	H7%	H8%	Gen.%
s42	.93	.132	87.1	0.0	6.5	7.5
s43	.13	—		*training collapse*		
s44	.98	.582	0.0	100	0.0	0.0
s45	.99	.506	5.1	93.9	1.0	1.0
s46	.93	.083	89.2	0.0	4.3	9.7

Table 1: Exec-only 4B GRPO at 500 steps (N=5). PR = prove rate, ISR* = mean ISR over proved samples, Gen. = genuine proofs. All converged seeds show ≥90% hacking.

Seed	Mode$_{500}$	ISR*$_{100}$	ISR*$_{500}$	ISR*$_{1000}$	Δ%
s42	H1	.642	.132	.003	−99.6
s45	H7→H1	.550	.506	.012	−97.6
s46	H1	.150	.083	.008	−94.7

Table 2: ISR* collapse trajectories over 1000 steps (N=3). Δ% = percent decrease from step 100 to step 1000. All seeds converge to ISR*<0.02.

4.2 Exec-Only Structural Degeneration

Table 1 presents per-seed results for 5 seeds of exec-only GRPO at 500 steps. All converged seeds (4/5; s43 suffers training collapse at PR=13%) achieve prove rates ≥93% while exhibiting ≥90% structural degeneration. The dominant hacking mode is seed-determined: s42 and s46 converge to H1 (conclusion embedding), while s44 and s45 converge to H7 (tautological shortcut). This bimodal attractor structure—stochastic bifurcation under identical hyperparameters—demonstrates that single-seed results are unreliable for characterizing GRPO behavior.

4.3 ISR Collapse Trajectories

Table 2 tracks ISR* across 100, 500, and 1000 training steps for three seeds extended to 1000 steps. All trajectories converge toward ISR*≈0, with collapse rates of 94.7–99.6%. Notably, s45 undergoes a mode transition from H7-dominant at 500 steps (ISR*=0.506) to H1-dominant at 1000 steps (ISR*=0.012), demonstrating that H7 is not a terminal attractor—continued training discovers the lower-ISR H1 exploit. This establishes that degeneration worsens monotonically with training: all roads lead to ISR≈0.

Seed	PR	H7%	Gen.%	Window?
s42	.81	16.0	17.3	✓
s43	.61	96.7	0.0	—
s44	.89	100	0.0	—
s45	.42	38.1	54.8	✓
s46	.77	14.3	54.5	✓
s47	.97	85.6	3.1	—
s48	.66	100	0.0	—
s49	.86	84.9	1.2	—
s50	.98	85.7	6.1	—
s51	.98	98.0	1.0	—

Table 3: ISR β=0.1 seed-level results (N=10, 4B). Genuine window defined as genuine rate >15%. Only 3/10 seeds sustain a genuine window.

N	ISR Gen.%	Rand. Gen.%	OR	p (CMH)
3	20.6	5.1	4.82	<0.001
5	22.6	7.7	3.48	$<10^{-8}$
8	13.9	6.9	2.18	0.008
10	11.3	7.3	1.62	0.015

Table 4: Pooled odds ratio for genuine proofs (ISR vs. random) as a function of sample size. OR shrinks monotonically; genuine window ratio drops from 2/3 to 3/10.

4.4 ISR as Reward: Goodhart Inevitability

We train 10 seeds with ISR as an auxiliary reward (β=0.1) and evaluate whether ISR opens a genuine reasoning window. Table 3 presents per-seed results. Only 3/10 seeds (s42, s45, s46) sustain a genuine window (genuine rate >15%), while the remaining 7 seeds collapse immediately to H7-dominated hacking (≥84.9% H7).

The genuine advantage of ISR over random diminishes as sample size increases (Table 4). At N=3, the pooled OR of 4.82 suggested a strong ISR benefit; at N=10, OR drops to 1.62 while remaining significant (CMH p=0.015). The genuine window ratio decreases from 2/3 to 3/10, establishing that the genuine window is the rare exception, not the modal outcome.

4.5 Cross-Scale Validation

Exec-only across scales. Table 5 compares exec-only GRPO across 1.7B, 4B, and 8B parameters. Goodhart collapse is universal—all scales show ≥93% hacking—but the dominant exploitation strategy shifts with capacity. At 1.7B, the model can only exploit via H7 (100% tautological shortcuts at PR=3%). At 4B, both H1 (51.6%) and H7 (48.5%) serve as attractors. At 8B, H8 (conjunction collapse) emerges as the primary mode (35.3%), displacing H7 (6.0%). This gradient—H7→H1/H7→H8—reflects a capacity-dependent minimum-effort hacking strategy: larger models access more sophisticated exploits because their optimization landscape contains stable basins at higher complexity levels.

Scale	n	PR	H1%	H7%	H8%	Gen.%
1.7B	1	.03	0	100	0	0
4B	4*	.96	51.6	48.5	3.0	4.6
8B	3	.84	40.3	6.0	35.3	7.0

Table 5: Cross-scale exec-only hacking mode distribution. ∗4B excludes s43 (training collapse). Dominant mode shifts from H7 (1.7B) to H1/H7 (4B) to H8 (8B).

Seed	PR	ISR*	H1	H7	H8	PG	Gen.
s42	.88	.148	25	5	52	5	8
s43	.73	.195	38	5	32	15	7
s44	.91	.075	58	8	22	—	6

Table 6: Qwen3-8B exec-only per-seed detail (%, 500 steps, K=8). H8 (conjunction collapse) is the primary or co-primary mode in 2/3 seeds, unlike 4B where H7 dominates.

Table 6 provides per-seed detail for 8B. The emergence of proof grafting (PG) as a minor mode (5–15%) at 8B—absent at 4B—further demonstrates that larger capacity enables a richer exploitation repertoire.

ISR across scales. Table 7 shows that ISR as a reward signal fails to prevent Goodhart collapse at any scale. The dominant hacking mode mirrors the exec-only pattern shifted by capacity: 1.7B collapses to H1 template gaming (100%, ISR=0.6 zero-variance), 4B to H7 tautology, and 8B to H8 conjunction collapse. The genuine window probability shows no clear scale trend (0%, 30%, 33%), suggesting it is a stochastic event that does not improve predictably with capacity.

4.6 Premise-Count Confound Analysis

ISR correlates strongly with premise count (Spearman ρ=0.67, $p < 10^{-10}$), and genuine proofs tend to use more premises than hacked proofs. Table 8 shows that after matching for premise count, the genuine OR drops from 3.48 to 1.12 (n.s., p=0.42), and the ISR quality mul-

Scale	n	PR	Dom. Hack	Window	Gen.%
1.7B	1	1.0	H1 template	0/1	0
4B	10	.76	H7 tautology	3/10	11.3
8B	3	.75	H8 conjunction	1/3	7.7

Table 7: Cross-scale ISR β=0.1 comparison. Modal outcome is collapse at all scales. Genuine window ratio does not improve with scale.

Metric	Unmatched	Matched	Interp.
Genuine OR	3.48	1.12	selection
ISR* ratio	1.06×	1.02×	no diff
ρ(ISR, #prem)	—	0.67	confound
Var. explained	—	45%	major

Table 8: Premise-count-matched analysis. ISR's genuine advantage is a selection effect: genuine proofs have more premises, which mechanically inflates ISR.

tiplier drops from 1.06× to 1.02×. Premise count alone explains 45% of ISR variance.

This reveals that the causal arrow runs from proof complexity to ISR, not from ISR to reasoning quality. ISR is a reliable *proxy* for proof complexity but does not independently measure reasoning quality. This finding has methodological implications beyond our setting: LOO ablation metrics in general require complexity control to avoid spurious quality signals.

4.7 Cross-Logic Transfer

STA transfers to Prolog (SWI-Prolog solver). Exec-only GRPO on Prolog achieves PR=53.3%±31.1% with ISR near-zero (2/3 seeds <0.04). STA analysis of 3 Prolog seeds reveals 3/3 gaming with 0% genuine: s42 shows clean H7 (100%), s43 degenerate syntax (97.4% H1), and s44 query-as-fact gaming (97.9% functional H7). The complete absence of genuine proofs in Prolog—compared to ∼6% residual in FOL—is consistent with Prolog's simpler semantics enabling more complete exploitation. This validates that STA's diagnostic capability generalizes across formal languages.

5 Discussion

Goodhart collapse is universal but strategy is capacity-dependent. Our cross-scale results establish a capacity-dependent exploitation gradient: 1.7B models are limited to the simplest exploits (H1 conclusion embedding, H9 contradiction injection), 4B models discover tautological shortcuts (H7) as a more efficient strategy, and 8B models access conjunction collapse (H8) as a yet more sophisticated exploit. This gradient has a natural interpretation: GRPO selects the minimum-effort hacking strategy available to each model's capacity. The KL penalty in GRPO makes strategies requiring less distributional shift more accessible—8B models, starting from a more capable distribution, find H8 closer in KL distance than H7, while 4B models must collapse further to the simpler H7 attractor.

The genuine window is the rare exception. At N=10, only 3 seeds sustain a genuine reasoning window under ISR reward. The shrinking OR (4.82→1.62) as sample size grows suggests that early small-sample estimates were inflated by favorable seed selection. The genuine window requires ISR to reinforce genuine proof production in early training before tautological exploitation is discovered—a narrow basin that most random initializations miss. This finding strengthens the Goodhart inevitability narrative: ISR occasionally opens a transient genuine window, but the modal outcome is collapse.

Stochastic bifurcation. The bimodal attractor structure in exec-only training (H1 vs. H7) and the 30% genuine window probability in ISR training both reflect stochastic bifurcation: identical hyperparameters produce qualitatively different outcomes depending on random seed. This has practical implications—single-seed evaluations of GRPO-based formal reasoning systems are unreliable, and multi-seed evaluation with structural diagnostics should be standard practice.

6 Conclusion

We introduced Solver Trace Ablation (STA) and the ISR metric as diagnostics for reward hacking in GRPO-trained formal reasoning. Our experiments across three model scales (1.7B, 4B, 8B), two logic types (FOL, Prolog), and multiple reward configurations establish three findings: (1) ISR reliably detects structural degeneration invisible to prove rate;

(2) ISR as a reward signal suffers inevitable Goodhart collapse, with genuine windows in only 30% of seeds and shrinking effect sizes at larger sample sizes; (3) exploitation strategy is capacity-dependent, with larger models discovering more sophisticated hacking modes rather than escaping Goodhart failure. The premise-count confound analysis further demonstrates that LOO ablation metrics require complexity control—a methodological contribution applicable beyond our specific setting.

Limitations

ISR measures syntactic necessity (whether removing a component breaks the proof) but not semantic faithfulness (whether the proof corresponds to intended reasoning). Spearman ρ(ISR, NLI accuracy) $= -0.049$ (p=0.187, n.s.), confirming ISR does not capture semantic quality. Our experiments use Qwen3 models at three scales; generalization to other model families (e.g., LLaMA) is not verified, though the 1.7B lower-capability bound experiment with LLaMA-3.2-3B (PR$\approx$15%) suggests the Goodhart phenomenon is not architecture-specific. The ISR β=0.05 no-safe-dose finding is based on 2 seeds; while both show deterministic collapse ($\geq$97% H7), larger sample sizes would strengthen this claim.

References

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Trung Quoc Luong, Xinbo Zhang, Zhanming Jie, Peng Sun, Xiaoran Jin, and Hang Li. 2024. ReFT: Reasoning with reinforced fine-tuning. In *Proceedings of ACL 2024*.

Huajian Xin, Daya Guo, Zhihong Shao, Zhizhou Ren, Qihao Zhu, Bo Liu, Chong Ruan, Wenda Li, and Xiaodan Liang. 2024. DeepSeek-Prover: Advancing theorem proving in LLMs through large-scale synthetic data. *arXiv preprint arXiv:2405.14333*.

Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. 2024. Solving Olympiad geometry without human demonstrations. *Nature*, 625:476–482.

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. 2016. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*.

Joar Skalse, Nikolaus HR Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and characterizing reward hacking. In *Advances in NeurIPS*, volume 35.

David Manheim and Scott Garrabrant. 2019. Categorizing variants of Goodhart's law. *arXiv preprint arXiv:1803.04585*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *Proceedings of ICML 2023*.

Thomas Coste, Usman Anwar, Robert Kirk, and David Krueger. 2024. Reward model ensembles help mitigate overoptimization. In *Proceedings of ICLR 2024*.

Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2016. Rationalizing neural predictions. In *Proceedings of EMNLP 2016*.

Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. 2020. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of ACL 2020*.

Albert Q. Jiang, Sean Welleck, Jin Peng Zhou, Wenda Li, Jia Liu, Mateja Jamnik, Timothée Lacroix, Yuhuai Wu, and Guillaume Lample. 2023. Draft, sketch, and prove: Guiding formal theorem provers with informal proofs. In *Proceedings of ICLR 2023*.

Emily First, Markus N. Rabe, Talia Ringer, and Yuriy Brun. 2023. Baldur: Whole-proof generation and repair with large language models. In *Proceedings of FSE 2023*.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.